

Safety Measures for Stormy Windows

by Bill York

In single-family homes, and many multifamily homes and condominiums, the existing, non-impact-rated windows and/or glass doors generally cannot be retrofitted to meet current impact standards and building codes; they can only be replaced or covered with impact-resistant products. But covering existing windows with shatter-resistant film may not offer the storm protection that is needed during harsh-wind events.

The use of safety and security films does reduce flying glass shards if the window glass is broken during a hurricane. Unfortunately, this is perhaps the least of your concerns. The truth is that some of the testing that has been done on safety and security films for storm protection is being misrepresented as meeting the impact standards for windows installed in the lower 30 ft of buildings. In these cases, the impact standards all call for large-missile testing and rating for a product to be used. The “large missile” is a 2 X 4 piece of lumber ranging in weight from 4 to 9 lb, depending on the design wind speed of the location where the product is to be used.

Many of the safety and security films are tested for small missile impacts, which consist of steel ball bearings striking the glass or product. These tests are reserved for windows that are located 30 or more ft above the ground only, where large-missile impacts are unlikely. In addition, the glass that is frequently used in the window film tests is $\frac{3}{16}$ inch thick tempered glass installed in a commercial

window frame—not what you usually find in most homes.

Therefore, the test conditions are not applicable for the windows installed in our homes, because the wrong test is being performed on a different type of window and glass. When homeowners have safety and security films installed on their windows, they may get a false sense of security and protection from storms.

Installation methods vary from installer to installer, and while some very specific installations may perform in a given storm, none of them meet the large-missile test standards that are required by the Florida Building Code.

The bottom line is this: To meet the current code in Florida in the windborne debris region, there are

two options for existing buildings. First option: The glazed openings could be protected with an impact resistant and rated product. Second option: The entire unit—frame and glass—would have to be replaced. Installation must be per the manufacturer's installation instructions and the product approval test report. That also means the building structure (rough opening) must be substantial enough for the new fenestration to be anchored adequately.

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Non-impact-rated windows need to be covered by plywood sheets in the event of a hurricane.

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